

SALES SIGHT 360

Multi-Channel Sales & Revenue Analytics Dashboard

INTERNSHIP PROJECT – 02

Data Analytics Project

 **Internship Organization :** Persevex

 **Intern :** Pratik Pradeep Patil

 **Department :** Computer Engineering

 **Internship Duration :** 3 Months

 **Project :** Project 02 – SalesSight 360

 **Technologies Used :**  Excel |  SQL |  Power Query |  DAX |  Power BI

Project Focus

Sales Performance • Revenue Analysis • Profitability • Forecasting

Project Workflow

Data Collection → Data Cleaning → Data Transformation → KPI Calculation → Power BI Dashboard → Business Insights

 **INTERNSHIP PROJECT DOCUMENTATION**

2026

TABLE OF CONTENTS

No.	Section
1	 Project Overview
2	 Business Problem
3	 Objectives
4	 Dataset Description
5	 Data Architecture
6	 Data Cleaning
7	 SQL Methodology
8	 Sales Definitions
9	 Forecasting Methodology
10	 KPI Definitions
11	 Power BI Dashboard
12	 Key Insights
13	 Recommendations
14	 Limitations
15	 Conclusion

1. PROJECT OVERVIEW

SalesSight 360 is a sales analytics project developed to analyze business sales data and present useful information through interactive dashboards. The project uses tools such as Microsoft Excel, SQL, and Power BI to organize, process, analyze, and visualize sales information.

The dashboard provides an overview of important business measures such as total revenue, total profit, number of orders, average order value, sales targets, and forecast values. It also allows users to explore sales performance by region, product category, sales channel, customer segment, and sales representative.

The main purpose of this project is to convert raw sales data into meaningful information that can help users understand business performance and make informed decisions.

2. BUSINESS PROBLEM

Businesses generate large amounts of sales data from different regions, products, customers, and sales channels. When this information is stored in separate files or viewed only in spreadsheet format, it may become difficult to understand overall sales performance.

Some common challenges are:

- Difficulty tracking total revenue and profit.
- Difficulty comparing sales performance across regions and channels.
- Limited visibility into product and category performance.
- Difficulty monitoring sales targets and expected revenue.
- Time-consuming manual reporting.
- Difficulty identifying sales trends and changes over time.

SalesSight 360 addresses these challenges by organizing sales information into a structured dataset and presenting it through interactive Power BI dashboards. Users can apply filters and compare different business areas to understand sales performance more easily.

3. OBJECTIVES

The main objectives of the SalesSight 360 project are:

1. Analyze sales performance: Calculate total revenue, profit, orders, and average order value.
 2. Understand regional performance: Compare sales across states and regions.
 3. Analyze product performance: Identify products and categories contributing to revenue and profit.
 4. Compare sales channels: Understand sales generated through different channels.
 5. Monitor sales targets: Compare forecast values with target revenue.
 6. Analyze the sales funnel: Review opportunities across different sales stages.
 7. Track sales representatives: Compare sales performance across representatives or teams.
 8. Create interactive dashboards: Provide slicers and filters for detailed analysis.
 9. Support decision-making: Present sales information in a clear and understandable format.
-

4. DATASET DESCRIPTION

The project uses structured sales data containing information about orders, customers, products, locations, sales channels, revenue, costs, and sales opportunities.

The dataset contains approximately 3,000 sales records. The exact number of records should be confirmed from the final dataset used in the project.

Main data fields

Field	Description
Order_ID	Unique identifier for an order
Order_Date	Date when the order was recorded
Customer_ID	Unique customer identifier
Customer_Name	Name of the customer

City	Customer or sales location city
State	State associated with the sale
Region	Business region
Customer_Segment	Customer group or segment
Product	Product name
Category	Product category
Quantity	Number of units sold
Unit_Price	Price per unit
Discount	Discount applied to the sale
Revenue	Sales revenue
Cost	Cost associated with the sale
Profit	Revenue after subtracting cost
Lead_Source	Source through which the lead was generated
Channel	Sales channel
Sales_Team	Team responsible for the sale
Sales_Rep	Sales representative
Sales_Stage	Current stage of the sales opportunity
Stage_Probability	Probability associated with the sales stage
Target_Revenue	Revenue target
Forecast_Value	Estimated future sales value
Variance_vs_Target	Difference between forecast and target
Order_Status	Current order status
Payment_Status	Payment status

Payment_Method	Method used for payment
----------------	-------------------------

Dataset purpose

These fields allow the project to analyze sales performance from different perspectives. For example, revenue can be compared by region, profit can be analyzed by category, and forecast values can be compared with targets.

5. DATA ARCHITECTURE

The project follows a simple data analytics workflow:

Excel/CSV → Power Query → SQL → Power BI

Step 1: Data collection

Sales data is collected from the provided Excel or CSV dataset. It contains information about customers, products, orders, sales stages, and financial values.

Step 2: Data cleaning

The dataset is checked for missing values, duplicate records, incorrect data types, inconsistent text, and formatting issues.

Step 3: Data storage and analysis using SQL

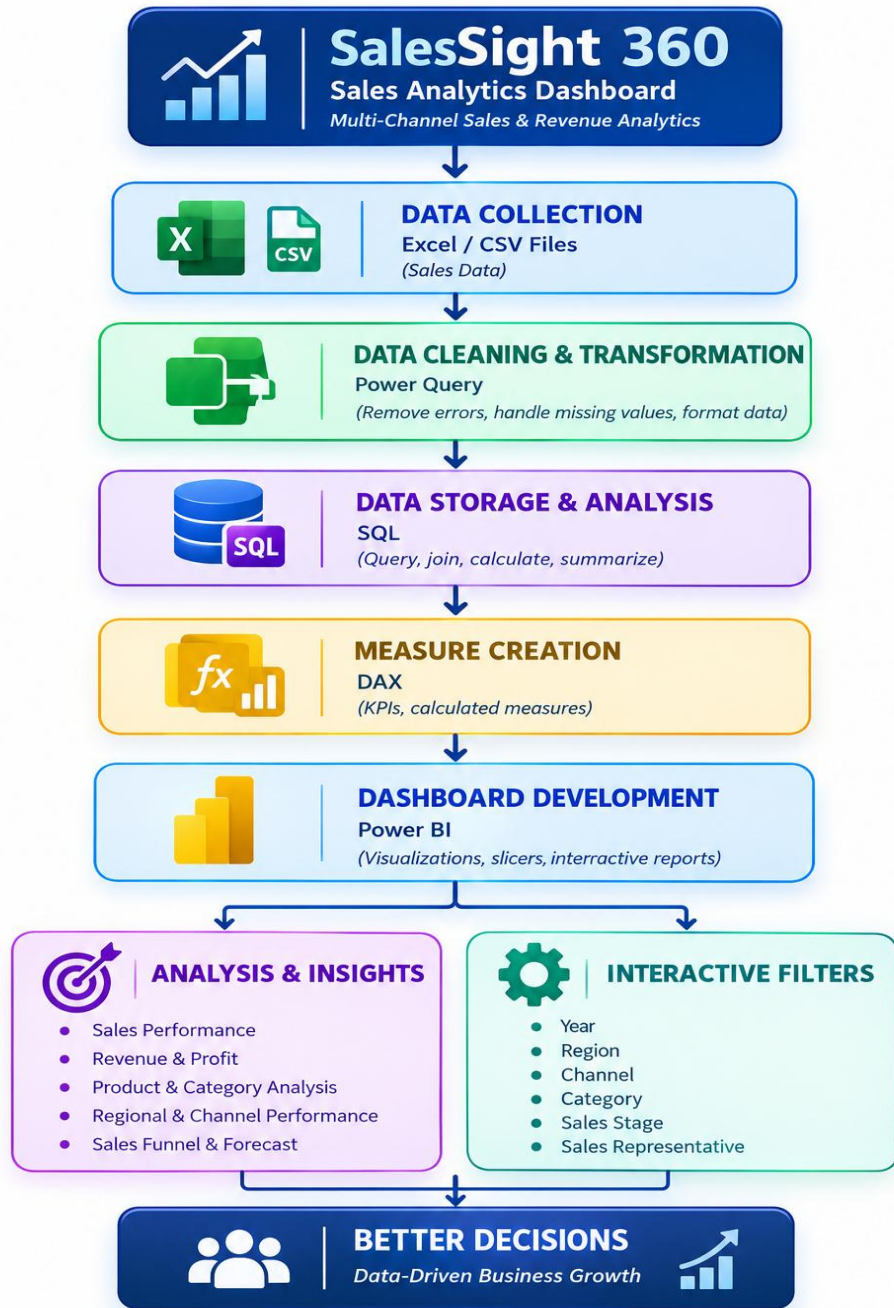
SQL is used to store, filter, join, and summarize the sales data. Queries can calculate revenue, profit, order counts, and performance by different categories.

Step 4: Data visualization using Power BI

The cleaned data is connected to Power BI. Measures, charts, cards, slicers, and tables are created to present sales performance.

Step 5: Dashboard analysis

Users interact with the dashboard by selecting filters such as year, region, channel, category, or sales representative. The visuals update according to the selected filters.



6. 🧹 DATA CLEANING

Data cleaning is an important step because incorrect or inconsistent data can affect the results of the analysis.

The following activities are performed or checked during data preparation:

6.1 Missing value handling

The dataset is checked for blank values in important fields such as order date, product, region, revenue, and profit. Missing values are handled according to their meaning and importance.

6.2 Duplicate record checking

Duplicate records are checked to avoid counting the same order or transaction more than once. Unique order identifiers are used where appropriate.

6.3 Data type correction

Columns are assigned suitable data types:

- Order_Date is stored as a date.
- Quantity is stored as a whole number.
- Revenue, Cost, Profit, and Forecast_Value are stored as numeric values.
- Names, categories, regions, and statuses are stored as text.

6.4 Text standardization

Text values are checked for unnecessary spaces, spelling differences, and inconsistent capitalization. For example, region names should use a consistent format.

6.5 Financial data validation

Revenue, cost, and profit values are checked for consistency. If profit is defined as revenue minus cost, the calculation should follow:

$$\text{Profit} = \text{Revenue} - \text{Cost}$$

6.6 Date preparation

Date fields are formatted consistently so that sales can be analyzed by year, quarter, month, day.

6.7 Data transformation

Power Query is used to prepare the data for analysis. Unnecessary columns may be removed, column names may be standardized, and suitable fields may be created for reporting.

7. SQL METHODOLOGY

SQL is used to manage and analyze structured sales data. It helps retrieve specific records, summarize information, and calculate business measures.

7.1 Data retrieval

SELECT statements are used to retrieve required columns and records from the sales table.

7.2 Filtering records

WHERE conditions are used to filter sales records based on region, category, channel, date, or sales stage.

7.3 Aggregation

Aggregate functions such as SUM(), COUNT(), and AVG() are used to calculate sales measures.

Examples include:

- Total revenue
- Total profit
- Number of orders
- Average order value
- Average sales value

7.4 Grouping data

GROUP BY is used to summarize sales by region, category, channel, product, or sales representative.

7.5 Joining tables

If the project uses separate customer, product, sales representative, or date tables, SQL joins can combine related information using common identifiers.

7.6 Business analysis

SQL queries help compare sales performance and prepare summarized data for further analysis in Power BI.

8. SALES DEFINITIONS

Revenue

Revenue is the total amount earned from sales before subtracting costs.

Profit

Profit is the amount remaining after subtracting the associated cost from revenue.

Order

An order represents a recorded customer purchase or sales transaction. Unique order IDs are used to count orders.

Average Order Value

Average Order Value (AOV) represents the average revenue generated per order.

$AOV = \text{Total Revenue} \div \text{Total Orders}$

Sales Target

A sales target is the revenue goal set for a specific period, team, representative, or business area.

Sales Stage

Sales stage represents the current status of a sales opportunity. Common stages include Lead, Qualified, Opportunity, Proposal, Negotiation, Closed Won, and Closed Lost.

Closed Won

Closed Won means the sales opportunity was successfully completed.

Closed Lost

Closed Lost means the sales opportunity was not completed successfully.

9. 🧠 FORECASTING METHODOLOGY

Sales forecasting estimates the sales value that may be achieved in the future. In SalesSight 360, forecast values can be used to compare expected sales with business targets.

9.1 Sales-stage probability

Different sales stages may have different probabilities of becoming successful sales. For example, an opportunity in the negotiation stage may have a higher probability than a newly generated lead.

The probabilities used in the project should be treated as assumptions unless they are calculated from historical business data.

9.2 Forecast value

A simplified weighted forecast can be calculated using:

Forecast Value = Opportunity Amount × Stage Probability

For example, if an opportunity amount is ₹50,000 and its stage probability is 60%, the weighted forecast value is ₹30,000.

This is an illustrative calculation, not a claim about the actual project results.

9.3 Forecast variance

Forecast variance shows the difference between forecast value and target revenue.

Forecast Variance = Forecast Value – Target Revenue

- Positive variance means forecast value is above the target.
- Negative variance means forecast value is below the target.
- Zero variance means forecast value equals the target.

9.4 Forecast limitations

Forecast values are estimates. Their accuracy depends on the quality of opportunity amounts, sales-stage probabilities, and source data.

10. 📊 KPI DEFINITIONS

KPIs (Key Performance Indicators) are measures used to monitor and evaluate business performance.

KPI	Definition	Formula / Calculation
💰 Total Revenue	The total sales revenue generated from all included sales records.	Sum of the Revenue column
📈 Total Profit	The total profit generated from the included sales records.	Sum of the Profit column
🛒 Total Orders	The number of unique orders in the dataset.	Distinct count of Order_ID
📄 Average Order Value (AOV)	The average revenue generated per order.	Total Revenue ÷ Total Orders
🌟 Forecast Value	The total expected sales value based on the forecast calculation used in the project.	Sum of the Forecast_Value column
🎯 Target Revenue	The revenue goal set for the selected period or business area. Targets should not be repeated at order level in a way that causes double counting.	Sum of target values at the correct target level
📊 Forecast Variance	The difference between forecast value and target revenue. A positive value means the forecast is above target; a negative value means it is below target.	Forecast Value – Target Revenue
🏆 Win Rate	The percentage of completed opportunities that were won. Open opportunities are excluded.	$\text{Closed Won} \div (\text{Closed Won} + \text{Closed Lost}) \times 100$

11. POWER BI DASHBOARD

The SalesSight 360 dashboard is designed to provide a clear view of sales performance through interactive visualizations.

Page 1: Executive Overview

This page presents a high-level summary of business performance.

KPI Cards

The page includes important measures such as:

- Total Revenue
- Total Profit
- Total Orders
- Average Order Value
- Forecast Value
- Win Rate

These cards help users quickly understand the selected data.

Revenue by State

This visual compares sales revenue across states. It helps users explore geographical sales performance.

Revenue by Channel

This chart compares revenue generated through different sales channels. It helps users understand how each channel contributes to sales.

Profit by Product Category

This visual compares profit across product categories and helps identify categories contributing to overall profitability.

Monthly Revenue vs Forecast

This line chart compares actual revenue with forecast values over time. It helps users review monthly sales trends and compare actual performance with expected sales.

Interactive Filters

Slicers such as Year, Region, Channel, and Category allow users to filter the dashboard and explore specific business areas.

Page 2: Sales Funnel & Forecast

This page focuses on sales opportunities, targets, forecasts, and representative performance.

Sales Funnel

The funnel visual displays the number or value of opportunities at different sales stages, such as Lead, Qualified, Proposal, Negotiation, Closed Won, and Closed Lost.

It helps users understand how opportunities are distributed across the sales process.

Target vs Forecast

This visual compares target revenue with forecast value. It helps users identify whether expected sales are above or below the selected target.

Forecast by Region

This chart compares forecast values across regions and helps users review expected sales performance geographically.

Closed Won by Lead Source

This visual compares successful opportunities by lead source. It helps users understand which sources are associated with completed sales.

Sales Representative Performance

A table can display sales representatives and selected measures such as revenue, profit, closed-won opportunities, and forecast value.

Interactive Filters

Filters such as Year, Month, Region, Sales Team, Sales Representative, and Lead Source allow users to explore the sales funnel and forecast data.

Page 3: Product & Customer Insights

This page provides a detailed view of product performance and customer purchasing patterns. It helps users understand which products and customer segments contribute to sales and profit.

Product Performance

This visual compares products based on revenue, profit, or quantity sold. It helps identify products that contribute more to business performance.

Revenue by Product Category

This chart displays total revenue across product categories. Users can compare categories and explore their contribution to overall sales.

Sales by Customer Segment

This visual compares revenue generated by different customer segments. It helps users understand which customer groups contribute to sales.

Profit by Product

This chart compares profit across individual products. It helps identify products with higher or lower profit contributions.

Quantity Sold by Product

This visual displays the quantity sold for different products. It helps users compare product demand within the selected filters.

Customer Location Analysis

A map or chart can display sales by customer location, such as state or city. It helps users explore geographical patterns in customer purchases.

Product and Customer Details

A table can display selected details such as product, category, customer segment, quantity sold, revenue, and profit. This allows users to review more specific information.

Interactive Filters

Slicers such as Year, Region, Category, Product, and Customer Segment allow users to filter the page and explore specific products or customer groups.

12. 💡 KEY INSIGHTS

The dashboard supports analysis of several business questions:

- Which states or regions generate higher revenue?
- Which sales channels contribute most to revenue?
- Which product categories generate higher profit?
- How does revenue change over time?
- How does actual revenue compare with forecast values?
- How are opportunities distributed across sales stages?
- Which regions have higher or lower forecast values?
- How does sales performance differ among representatives?

The actual answers should be written after reviewing the final dashboard results. Avoid adding specific findings unless they are supported by the dataset and visuals.

13. 🎯 RECOMMENDATIONS

Based on the analysis performed, businesses can consider the following actions:

📍 Review regional performance

Compare revenue and profit across regions to understand differences in sales performance.

🛒 Monitor product categories

Review category-level revenue and profit to understand which products contribute to business results.

📱 Track sales channels

Compare channels regularly to monitor their contribution to total sales.

Monitor targets

Review forecast variance to identify periods or areas where expected sales are below target.

Review sales opportunities

Monitor open opportunities and their stages to understand the current sales pipeline.

Support sales representatives

Use representative-level reports to review performance and identify areas where additional support may be useful.

Use regular reporting

Refresh the dashboard with updated data and review KPIs regularly to monitor changes over time.

14. LIMITATIONS

Although SalesSight 360 provides useful sales analysis, the project has some limitations:

1. Data quality: Results depend on the accuracy and completeness of the dataset.
 2. Forecast assumptions: Forecast values depend on the opportunity amounts and stage probabilities used.
 3. Historical information: If the dataset does not store previous sales-stage changes, it cannot fully show how opportunities moved through the funnel over time.
 4. Target structure: Target values must be stored at a suitable level to avoid counting the same target multiple times.
 5. Dataset scope: Findings apply only to the data included in the project.
 6. Forecast uncertainty: Forecasts are estimates and may differ from actual future sales.
 7. Business context: The dashboard provides analysis, but it does not automatically explain every reason behind changes in sales performance.
-

15. CONCLUSION

SalesSight 360 – Sales Analytics Dashboard demonstrates how Excel, SQL, and Power BI can be used to prepare, analyze, and visualize sales data.

The project combines sales information into an interactive dashboard that presents revenue, profit, orders, product performance, regional performance, sales channels, sales stages, and forecasts.

The Executive Overview provides a summary of sales performance, while the Sales Funnel & Forecast page focuses on opportunities, targets, and expected sales.

Overall, the project provides practical experience in data cleaning, SQL analysis, KPI development, data visualization, and business reporting. It demonstrates how data analytics tools can transform raw sales records into clear and useful business information.

Tools and Technologies

-  **Microsoft Excel:** Data storage and initial review
-  **Power Query:** Data cleaning and transformation
-  **SQL / MySQL:** Data querying and analysis
-  **Microsoft Power BI:** Dashboard development and visualization
-  **DAX:** KPI and measure calculations

References

- Project sales dataset and supporting files
 - Microsoft Excel documentation
 - Microsoft SQL / MySQL documentation
 - Microsoft Power BI documentation
 - Project dashboard and implemented SQL queries
-